

Predicting treatment delays of more than 90 days in prostate cancer: added value of county rurality, county income and marital status over clinical characteristics

A machine learning cohort study of 330,827 men treated with surgery or radiotherapy in US SEER registries, 2010 to 2022

Tanmay Hinge Code, protocol and full analysis reports: <https://github.com/tanmayhinge/seer-prostate>

How to read this supplement

- **Source of tables.** Every table is generated by script from the analysis reports in the public repository, so the values match the analysis outputs. The reports, in the `reports` folder, contain further tables not reproduced here, including results for every model step, leave-one-variable-out refits, bounds by income and marital status, and the protocol amendment log.
- **Skill and intervals.** Skill added is in percentage points of out-of-fold log-loss skill. Intervals are 95% cluster bootstrap intervals over rurality by county income cells (500 resamples of out-of-fold predictions, no refitting) unless stated, so they exclude model-fitting variability.
- **Disclosure control.** As the SEER Research Data Use Agreement requires, statistics resting on 1 to 4 men are shown as <5, and numbers of men in cross-tabulations are rounded to the nearest 10.
- **Post-review analyses.** Tables S3 and S4 and the indices by period in Table S7 come from analyses specified after an internal review and before they were run (protocol amendment 1.9). The two-model agreement rule for standardised contrasts was adopted post hoc (amendment 1.7).

Box S1. Survival comparisons and immortal time bias (not estimated)

Comparing survival between treated and untreated men is exposed to immortal time bias, because treated men must survive until treatment starts. A simulation study of observational research found that time-fixed and exclusion methods overestimated a treatment's benefit, that a 1-year landmark method reduced the bias but did not remove it, and recommended the time-dependent method [1]. A suitable design would treat treatment as a time-dependent exposure, with competing risks of prostate cancer death and death from other causes, and use a 12-month landmark analysis only as a sensitivity analysis. This design is described, not estimated, in this study.

Box S2. Candidate Australian analogues of the SEER variables (no Australian data analysed)

Published analyses of the Prostate Cancer Outcomes Registry in Tasmania used the measures below [2, 3]. They are candidate analogues, not equivalents: the geographic units, the construction of the measures and the health systems differ, and availability for any new study must be confirmed against the registry's data dictionary.

- **Rurality.** The county Rural-Urban Continuum Code has a candidate analogue in Australian Statistical Geography Standard remoteness areas, assigned by residential postcode.
- **Area income.** County median household income has a candidate analogue in the SEIFA Index of Relative Socio-Economic Advantage and Disadvantage, assigned by postcode.
- **Insurance and sector.** SEER has no insurance variable. The closest Australian contrast studied is a public or private treating facility.

Data gaps reported in those analyses were no comorbidity data [2], patient-reported outcomes collected consistently only from 2018 [3], and no way to tell whether external beam radiotherapy was given in a public or private facility [3]. Referral, biopsy and multidisciplinary meeting dates are not assumed to be available.

Supplementary tables

Table S1. Crude waiting beyond 90 days by risk group and county rurality

Risk group	County rurality	Men	Waited more than 90 days (%)	Median days	90th percentile days
Low	Metro, 1 million or more	34,260	50.7	91	217
	Metro, 250,000 to 1 million	11,930	47.0	86	211
	Metro, under 250,000	4,780	42.3	82	198.2
	Nonmetro, adjacent to metro	3,990	39.0	77	187
	Nonmetro, not adjacent to metro	2,510	39.0	76	177
	Unknown	20	34.8	81	122.8
Intermediate	Metro, 1 million or more	75,190	45.5	84	182
	Metro, 250,000 to 1 million	28,140	41.1	79	172
	Metro, under 250,000	10,070	36.7	75	165
	Nonmetro, adjacent to metro	8,830	36.1	74	161
	Nonmetro, not adjacent to metro	5,360	36.1	71	160
	Unknown	40	42.5	71	200.5
High	Metro, 1 million or more	75,760	35.7	73	157
	Metro, 250,000 to 1 million	29,320	30.0	67	142
	Metro, under 250,000	10,130	27.9	65	140
	Nonmetro, adjacent to metro	9,590	27.7	63	134.5
	Nonmetro, not adjacent to metro	6,010	26.0	62	133
	Unknown	90	28.4	61.5	141.5
Unknown	Metro, 1 million or more	9,460	44.2	82	224
	Metro, 250,000 to 1 million	2,770	42.2	80	203
	Metro, under 250,000	1,120	35.0	70	179
	Nonmetro, adjacent to metro	950	36.5	72	174.2
	Nonmetro, not adjacent to metro	510	29.6	63.5	162
	Unknown	10	<5	77	212.7

Crude values; nothing is adjusted. Numbers of men are rounded to the nearest 10. Medians and 90th percentiles count top-coded intervals as 731 days. <5: statistic suppressed because 1 to 4 men in the row did, or did not, wait more than 90 days.

Table S2. Performance at step 2 by risk group and model

Stratum	Model	Log-loss skill (%)	Brier skill (%)	AUC	Calibration intercept	Calibration slope
All men	Logistic	3.63	4.66	0.638	−0.000	1.000
	LightGBM	4.17	5.34	0.648	−0.000	1.030
Low risk	Logistic	0.76	1.03	0.592	0.000	0.988
	LightGBM	0.99	1.35	0.604	−0.000	0.937
Intermediate risk	Logistic	1.06	1.40	0.605	−0.000	1.001
	LightGBM	1.48	1.96	0.615	−0.000	1.008
High risk	Logistic	4.39	5.42	0.654	0.000	1.001
	LightGBM	4.90	6.04	0.663	−0.000	1.023
Unknown risk	Logistic	1.07	1.43	0.610	0.000	0.982
	LightGBM	1.96	2.64	0.629	−0.000	0.882

Out-of-fold predictions at step 2 (clinical need plus area and marital characteristics). Skill is against the step 0 model. Calibration intercept: 0 is ideal; slope: 1 is ideal, below 1 means predictions are too extreme. Values for every step are in `reports/phase4_models.md`.

Table S3. Robustness of the area and marital increment to tuning, fold assignment and clustering

Stratum	Model	Primary analysis (tuned at step 3)	Tuned at each step	Mean (range) over 10 fold seeds	Clustered by county income band only
All men	Logistic	0.62 (0.39 to 0.87)	0.62 (0.39 to 0.87)	0.62 (0.61 to 0.62)	0.62 (0.38 to 0.90)
	LightGBM	0.99 (0.73 to 1.30)	0.99 (0.73 to 1.30)	0.99 (0.98 to 0.99)	0.99 (0.72 to 1.35)
Low risk	Logistic	0.62 (0.32 to 0.94)	0.62 (0.32 to 0.94)	0.62 (0.59 to 0.63)	0.62 (0.28 to 1.07)
	LightGBM	0.96 (0.67 to 1.28)	0.96 (0.67 to 1.28)	0.98 (0.95 to 1.01)	0.96 (0.64 to 1.36)
Intermediate risk	Logistic	0.56 (0.35 to 0.83)	0.56 (0.34 to 0.83)	0.56 (0.54 to 0.56)	0.56 (0.33 to 0.89)
	LightGBM	0.93 (0.65 to 1.27)	0.93 (0.65 to 1.27)	0.93 (0.91 to 0.97)	0.93 (0.63 to 1.31)
High risk	Logistic	0.86 (0.60 to 1.15)	0.86 (0.60 to 1.15)	0.85 (0.85 to 0.86)	0.86 (0.58 to 1.15)
	LightGBM	1.18 (0.90 to 1.56)	1.18 (0.90 to 1.54)	1.18 (1.16 to 1.20)	1.18 (0.87 to 1.55)
Unknown risk	Logistic	0.42 (0.08 to 0.80)	0.40 (0.07 to 0.80)	0.41 (0.32 to 0.48)	0.40 (0.07 to 0.78)
	LightGBM	1.37 (0.80 to 2.19)	1.37 (0.80 to 2.19)	1.29 (1.22 to 1.43)	1.37 (0.78 to 2.09)

Percentage points of out-of-fold log-loss skill added by area and marital characteristics (step 1 to 2), with 95% cluster bootstrap intervals (first fold seed, 500 resamples, no refitting). Tuning was not repeated per seed. Specified after an internal review (protocol amendment 1.9).

Table S4. Stage-free clinical block and year of diagnosis as categories

Analysis	Stratum	Model	Clinical need added (95% interval)	Area and marital added (95% interval)
Stage-free clinical block	All men	Logistic	2.97 (2.64 to 3.34)	0.62 (0.39 to 0.87)
		LightGBM	3.13 (2.80 to 3.48)	1.02 (0.75 to 1.33)
	Low risk	Logistic	0.14 (0.08 to 0.20)	0.62 (0.32 to 0.94)
		LightGBM	0.02 (−0.09 to 0.13)	0.96 (0.67 to 1.28)
	Intermediate risk	Logistic	0.49 (0.41 to 0.57)	0.56 (0.34 to 0.83)
		LightGBM	0.55 (0.44 to 0.65)	0.93 (0.65 to 1.27)
	High risk	Logistic	3.42 (3.07 to 3.76)	0.85 (0.59 to 1.14)
		LightGBM	3.64 (3.27 to 3.99)	1.19 (0.91 to 1.57)
	Unknown risk	Logistic	0.66 (0.41 to 0.92)	0.40 (0.06 to 0.80)
		LightGBM	0.59 (0.27 to 0.95)	1.37 (0.80 to 2.19)
Year as categories	All men	Logistic	2.98 (2.66 to 3.34)	0.62 (0.39 to 0.86)
	Low risk	Logistic	0.15 (0.09 to 0.21)	0.60 (0.31 to 0.92)
	Intermediate risk	Logistic	0.50 (0.42 to 0.58)	0.56 (0.34 to 0.83)
	High risk	Logistic	3.48 (3.12 to 3.82)	0.87 (0.61 to 1.16)
	Unknown risk	Logistic	0.65 (0.41 to 0.90)	0.47 (0.11 to 0.89)

Stage-free: summary stage removed from the clinical block, risk strata unchanged. Year as categories: logistic regression only, one indicator per year replacing linear year and the 2020 indicator. Both use per-step tuning and the first fold seed. Primary estimates are in Table 2 of the main text. Specified after an internal review (protocol amendment 1.9).

Table S5. Sensitivity analyses

Scenario	Threshold (days)	Men	Delayed (%)	Added, logistic (95% interval)	Added, LightGBM (95% interval)	Area contrast	Never married contrast
Primary analysis	90	330,830	39.7	0.62 (0.39 to 0.87)	0.99 (0.73 to 1.30)	−9.2 / −8.8	6.7 / 5.9
Threshold of 60 days	60	330,830	66.6	0.54 (0.31 to 0.79)	0.94 (0.73 to 1.18)	−10.6 / −10.3	3.7 / 3.2
Threshold of 120 days	120	330,830	23.0	0.68 (0.45 to 0.93)	1.06 (0.79 to 1.38)	−6.9 / −7.1	6.0 / 5.0
Threshold of 180 days	180	330,830	9.1	0.66 (0.41 to 0.92)	1.00 (0.69 to 1.35)	−3.0 / −3.2 ^a	3.4 / 3.1
Risk groups from Gleason score and PSA only	90	330,830	39.7	0.62 (0.39 to 0.87)	0.99 (0.73 to 1.30)	−9.2 / −8.8	6.7 / 5.9
Radical prostatectomy without radiotherapy only	90	159,580	41.1	0.66 (0.48 to 0.88)	1.00 (0.79 to 1.26)	−8.9 / −7.8	7.1 / 6.9
Excluding men diagnosed in 2020	90	306,960	39.6	0.64 (0.40 to 0.91)	1.03 (0.75 to 1.35)	−9.4 / −8.9	6.8 / 5.9
Including men diagnosed in 2023	90	358,540	40.8	0.56 (0.35 to 0.80)	0.94 (0.69 to 1.22)	−8.6 / −8.3	6.4 / 5.8
Including intervals of 0 days	90	338,580	38.8	0.61 (0.38 to 0.86)	0.97 (0.70 to 1.27)	−9.2 / −9.2	6.5 / 5.8
One primary cancer only (strict first primary)	90	296,120	40.1	0.61 (0.38 to 0.85)	0.98 (0.73 to 1.27)	−9.1 / −9.3	6.6 / 5.6
Excluding prostatectomy not otherwise specified	90	330,180	39.7	0.62 (0.39 to 0.87)	0.98 (0.72 to 1.28)	−9.2 / −9.1	6.7 / 5.9

All men. Each scenario changes one setting from the primary analysis. Added: percentage points of log-loss skill added by area and marital characteristics. Contrasts are standardised differences in percentage points (logistic / LightGBM): nonmetro counties not adjacent to a metro area minus metro areas of 1 million or more, each at its typical county income; and never married minus married. ^aTwo-model rule (both 3 points or more, same direction) not met. Threshold scenarios use a different outcome, so their values are not directly comparable. Numbers of men are rounded to the nearest 10.

Table S6. Standardised contrasts in waiting more than 90 days, by risk group

Stratum	Contrast	Logistic	LightGBM	Two-model rule met
All men	Observed minus reference profile	−1.1	−1.6	no, both under 3
	Nonmetro not adjacent minus metro 1 million or more	−9.2	−8.8	yes
	Never married minus married	6.7	5.9	yes
Low risk	Observed minus reference profile	−3.4	−4.7	yes
	Nonmetro not adjacent minus metro 1 million or more	−11.3	−13.5	yes
	Never married minus married	3.6	2.4	no, one model only
Intermediate risk	Observed minus reference profile	−1.4	−1.6	no, both under 3
	Nonmetro not adjacent minus metro 1 million or more	−8.7	−9.5	yes
	Never married minus married	6.4	5.5	yes
High risk	Observed minus reference profile	0.1	−0.1	no, both under 3
	Nonmetro not adjacent minus metro 1 million or more	−8.4	−7.2	yes
	Never married minus married	8.3	7.6	yes
Unknown risk	Observed minus reference profile	−0.1	0.0	no, both under 3
	Nonmetro not adjacent minus metro 1 million or more	−12.1	−14.2	yes
	Never married minus married	6.0	6.2	yes

Standardised percentage points waiting more than 90 days (g-computation with the step 2 model; each man keeps his own clinical features and year). Reference profile: married, metro area of 1 million or more, county income rank at the 83.33% quantile. Area profiles set rurality and county income together at the median income band of men living in that type of area. Rule (post hoc, amendment 1.7): both models 3 points or more in the same direction. No intervals are reported.

Table S7. Income concentration of delay, overall and by period of diagnosis

Stratum	2010 to 2022	2010 to 2014	2015 to 2019	2020 to 2022
All men	0.062 (0.039 to 0.083)	0.059 (0.025 to 0.092)	0.037 (0.011 to 0.065)	0.027 (0.003 to 0.061)
Low risk	0.103 (0.068 to 0.134)	0.092 (0.053 to 0.136)	0.084 (0.052 to 0.121)	0.052 (0.017 to 0.092)
Intermediate risk	0.075 (0.051 to 0.096)	0.058 (0.023 to 0.099)	0.048 (0.019 to 0.080)	0.039 (0.015 to 0.067)
High risk	0.041 (0.017 to 0.065)	0.028 (−0.011 to 0.062)	0.020 (−0.004 to 0.049)	0.015 (−0.013 to 0.056)
Unknown risk	0.077 (0.037 to 0.128)	0.060 (0.001 to 0.130)	0.023 (−0.041 to 0.100)	0.049 (−0.005 to 0.101)

Erreygers-corrected concentration index (95% cluster bootstrap interval) of waiting more than 90 days, ranking men by county median household income, poorest first. Positive values mean delay is concentrated in higher-income counties. Crude: not standardised for clinical features, and it does not separate income from rurality. Indices by period were specified after an internal review (amendment 1.9).

Table S8. Men without a recorded interval: bounds and inverse probability weighting

County rurality	Treated men	Recorded men (%)	Lower bound (%)	Upper bound (%)	No recorded interval (%)	Weighted (%)
Metro, 1 million or more	205,980	42.5	40.2	45.7	5.5	42.3
Metro, 250,000 to 1 million	75,790	37.6	35.8	40.6	4.8	37.4
Metro, under 250,000	27,050	34.2	33.0	36.6	3.6	34.1
Nonmetro, adjacent to metro	24,050	33.2	32.2	35.1	2.9	33.1
Nonmetro, not adjacent to metro	14,780	32.2	31.3	34.0	2.7	32.0
Unknown	170	33.3	31.9	36.1	4.2	33.5

Percentage waiting more than 90 days among treated men meeting cohort steps 1 to 6 (intervals of 0 days excluded). Lower bound: every man without a recorded interval waited 90 days or less; upper bound: every such man waited longer. Weighted: men with a recorded interval, weighted by the inverse of their predicted probability of having one. Numbers of men rounded to the nearest 10. Neither approach addresses selection into recorded treatment.

Table S9. Secondary outcome: recorded radical prostatectomy or radiotherapy

Stratum	Model	Skill at step 2 (%)	Clinical need added (95% interval)	Area and marital added (95% interval)	AUC at step 2	Area contrast	Never married contrast
Intermediate and high risk	Logistic	17.18	15.22 (14.85 to 15.65)	1.96 (1.72 to 2.31)	0.771	−2.9	−6.5
	LightGBM	20.49	18.32 (17.92 to 18.77)	2.16 (1.92 to 2.54)	0.791	−1.4	−5.9
Intermediate risk	Logistic	8.79	6.81 (6.38 to 7.21)	1.97 (1.65 to 2.42)	0.692	−2.2	−7.2
	LightGBM	10.14	7.83 (7.42 to 8.20)	2.31 (1.99 to 2.82)	0.703	−0.3	−6.4
High risk	Logistic	29.29	27.25 (26.63 to 27.96)	2.04 (1.84 to 2.33)	0.854	−3.3	−5.8
	LightGBM	31.76	29.54 (28.94 to 30.17)	2.22 (2.00 to 2.53)	0.865	−2.5	−5.1

Outcome: a record of radical prostatectomy or radiotherapy in the first course, among intermediate- and high-risk men meeting every cohort step before treatment. No record can mean active surveillance, watchful waiting, hormone therapy only, refusal or uncaptured treatment, so a lower percentage cannot be read as under-treatment. Contrasts are standardised percentage points, defined as in Table S6.

Supplementary figures

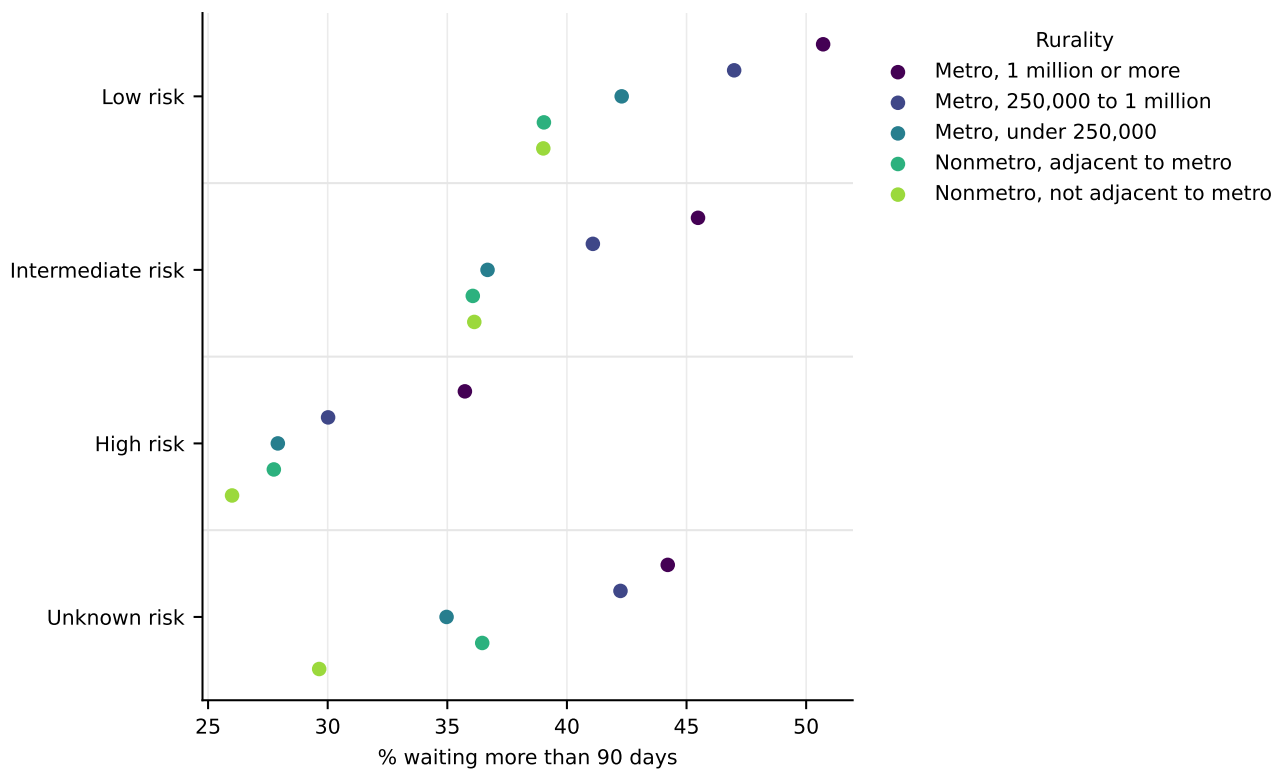


Figure S1. Crude percentage of men waiting more than 90 days from diagnosis to first recorded treatment, by risk group and county rurality. Men with unknown rurality, and percentages resting on 1 to 4 men, are not shown. Nothing is adjusted.

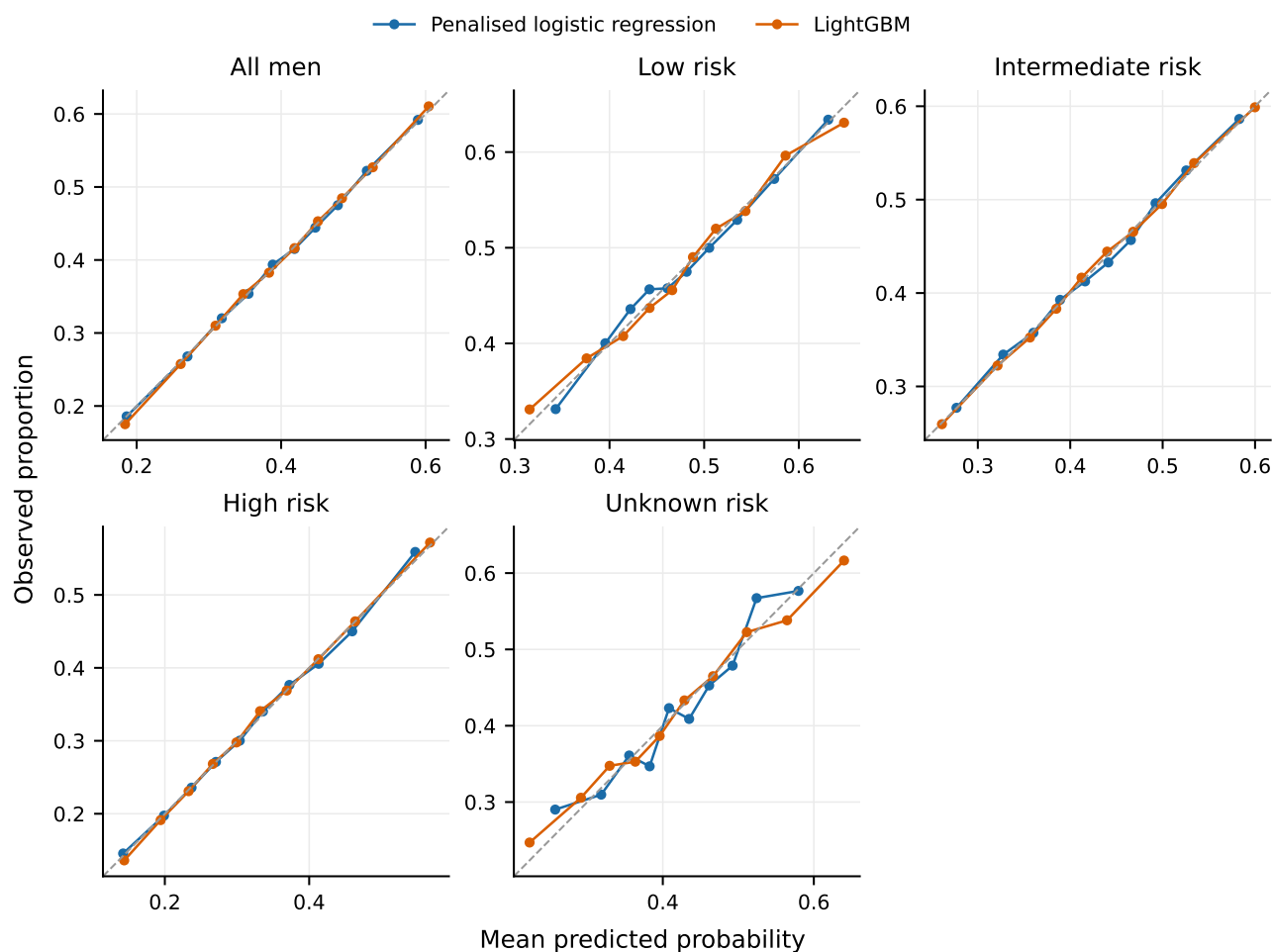


Figure S2. Calibration of out-of-fold predicted probabilities at step 2 (clinical need plus area and marital characteristics), in 10 equal-count bins, by risk group and model type. The dashed line marks perfect calibration.

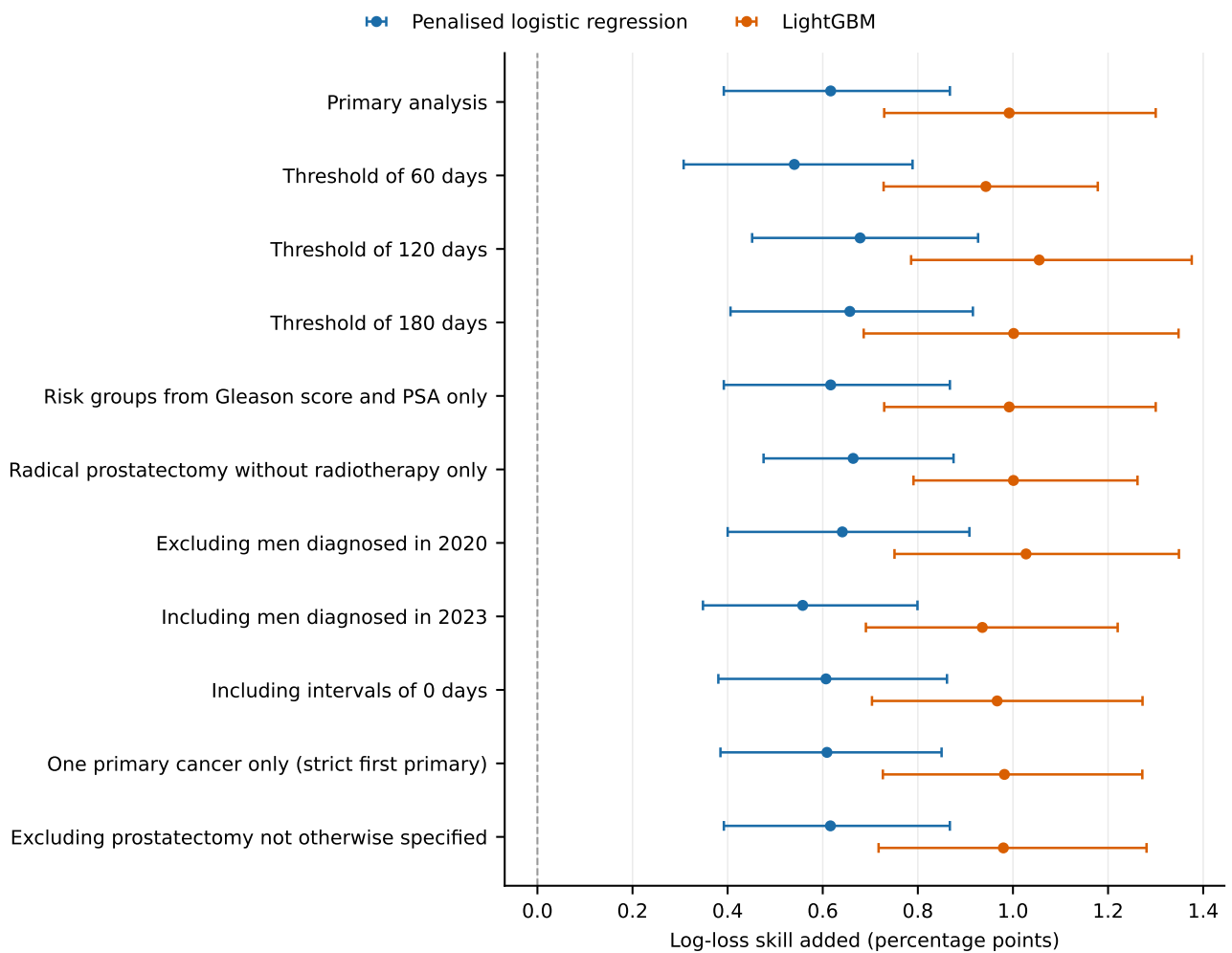


Figure S3. Log-loss skill added by the area and marital block (labelled social position in the figure) under each sensitivity scenario, all men, with 95% cluster bootstrap intervals. Each scenario changes one setting from the primary analysis. The threshold scenarios use a different outcome, so their values are not directly comparable with the others.

Reporting checklists

Table S10. STROBE and RECORD checklist. Item wording is abbreviated from the RECORD statement, which includes the STROBE items [4, 5]. Locations are sections of the main text unless marked S (this supplement).

Item	Topic	Where reported
1a, 1b	Design in title or abstract; informative abstract	Abstract (retrospective cohort study)
RECORD 1.1 to 1.3	Database, region and time frame; linkage	Title; Abstract; Methods, Data and cohort (county attributes linked by SEER; no person-level linkage)
2, 3	Background; objectives	Introduction
4, 5	Study design; setting and dates	Methods, Data and cohort
6a	Eligibility and selection	Methods, Data and cohort; Figure 1
6b	Matching	Not applicable
RECORD 6.1 to 6.3	Selection codes; their validation; linkage flow	Methods, Data and cohort (codes in <code>config/analysis.yaml</code>); no validation study of the cohort algorithm; no person-level linkage
7, RECORD 7.1	Outcomes and predictors; complete list of codes	Methods, Outcome and risk groups, and Ordered feature steps; <code>config/analysis.yaml</code>
8	Data sources and measurement	Methods, Data and cohort, and Outcome and risk groups
9	Efforts to address bias	Methods, Robustness, selection and secondary outcome; Discussion (limitations)
10	Study size	Methods, Data and cohort (all eligible men); Figure 1
11	Quantitative variables	Methods, Ordered feature steps
12a, 12b	Statistical methods; subgroups and interactions	Methods, Models and validation (risk-group strata; clinical interaction terms), and Standardised percentages and income concentration
12c	Missing data	Methods, Ordered feature steps, Models and validation, and Robustness, selection and secondary outcome (bounds and weighting); Table S8
12d	Loss to follow-up	Not applicable (top-coded intervals counted as delayed)
12e	Sensitivity analyses	Methods, Robustness, selection and secondary outcome; Tables S3 to S5
RECORD 12.1 to 12.3	Database access; data cleaning; linkage	Full case listing of the selected site and years; every export label mapped explicitly in code; no person-level linkage
13a to 13c, RECORD 13.1	Numbers at each stage; flow diagram	Results, Cohort and delay; Figure 1
14a, 14b	Characteristics; missing data	Table 1 (unknown categories shown)
14c	Follow-up time	Not applicable
15	Outcome data	Results, Cohort and delay; Table S1
16a	Estimates and precision	Table 2 (intervals); standardised contrasts without intervals, stated in Results; Table S6
16b, 16c	Category boundaries; absolute measures	Methods, Outcome and risk groups; Table 1; Results, Standardised differences and income concentration
17	Other analyses	Results, Robustness, selection and secondary outcome; Tables S3 to S9
18, 20	Key results; interpretation	Discussion
19, RECORD 19.1	Limitations, including routinely collected data	Discussion, limitations paragraph
21	Generalisability	Discussion
22	Funding	Declarations
RECORD 22.1	Access to protocol, data and code	Methods, Protocol, reporting and disclosure; Declarations

Table S11. TRIPOD+AI checklist. Item wording is abbreviated from the TRIPOD+AI checklist [6]. Development and internal evaluation use the same data through cross-fitting; the models are measurement tools and are not proposed for clinical use.

Item	Topic	Where reported
1, 2	Title; abstract	Title; Abstract
3a to 3c, 4	Context; intended purpose; known inequalities; objectives	Introduction; Methods, Protocol, reporting and disclosure (not for clinical use)
5a, 5b, 6a to 6c	Data sources, dates, setting, eligibility, treatments	Methods, Data and cohort, and Ordered feature steps (treatment type at step 3)
7	Data preparation	Code in the repository
8a to 8c	Outcome; assessors; blinding	Methods, Outcome and risk groups; assessors and blinding not applicable (registry-abstracted outcome)
9a to 9c	Predictors	Methods, Ordered feature steps (pre-specified in the protocol); assessors not applicable

Item	Topic	Where reported
10, 11	Sample size; missing data	Methods, Data and cohort (all eligible men), Ordered feature steps, and Models and validation
12a to 12c	Data use; predictor handling; model type, tuning and internal validation	Methods, Models and validation (5-fold cross-fitting)
12d, 23b	Heterogeneity across clusters	Cluster bootstrap only; heterogeneity across clusters not examined
12e	Performance measures	Methods, Models and validation; Table S2; Figure S2
12f, 24	Model updating	Not applicable
12g	How predictions were calculated	Code in the repository
13	Class imbalance	No class imbalance methods used
14	Fairness	Not addressed as a model property; area and marital differences are the measured quantity
15	Model output	Predicted probabilities; no classification threshold
16	Training versus evaluation differences	Not applicable (cross-fitting within the same data)
17, 18a to 18f, 19	Ethics; funding; conflicts; protocol; registration; data and code; patient involvement	Methods, Protocol, reporting and disclosure (not registered); Declarations
20a, 20b	Participant flow; characteristics	Results, Cohort and delay; Figure 1; Table 1
20c	Comparison with development data	Not applicable
21	Participants and events per analysis	Table 1; Table 2; Table S1
22	Full model specification	Not released as a clinical model; code that refits every model is in the repository
23a	Performance with intervals, including subgroups	Table 2 and Table S2 by risk group (AUC and calibration without intervals); not reported by area or marital subgroup
25, 26	Interpretation; limitations	Discussion
27a to 27c	Implementation; next steps	Not applicable (no implementation proposed); Discussion

References

- [1] Zheng Q, Otahal P, Cox IA, et al. The influence of immortal time bias in observational studies examining associations of antifibrotic therapy with survival in idiopathic pulmonary fibrosis: a simulation study. *Front Med (Lausanne)*. 2023;10:1157706.
- [2] Foley GR, Blizzard CL, Stokes B, et al. Urban-rural prostate cancer disparities in a regional state of Australia. *Sci Rep*. 2022;12(1):3022.
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- [6] Collins GS, Moons KGM, Dhiman P, et al. TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods. *BMJ*. 2024;385:e078378.